Domain Adaptive Remote Sensing Scene Recognition via Transferable Discriminative Visual Prompting

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***Abstract*—The task of remote sensing (RS) scene recognition suffers from significant distribution shifts induced by complex and variable observation environments, making unsupervised domain adaptation (UDA) a promising solution. However, existing prompt-based UDA methods mainly focus on token-level alignment within either modality and often overlook structural adaptability and spatially discriminative regions (DRs). In this article, we propose a novel framework termed domain adaptive RS scene recognition via transferable discriminative visual prompting (TDVP) to overcome these limitations. In particular, we introduce the DR guidance module (DRGM), an architectural enhancement designed to inject region-specific discriminative representations directly into the vision encoder. Built upon this foundation is a learnable cross-domain semantic prompt (LCSP), which leverages a region-aware mask to generate spatially adaptive representations that facilitate semantic alignment and enhance transferability across domains. Finally, we introduce prompt consistency regularization (PCR), which enforces semantic consistency between visual representations obtained with and without the region-aware mask, thereby enhancing the stability and generalizability of learned prompts across domains. Extensive experiments demonstrate that our approach effectively addresses domain shifts and consistently outperforms existing state-of-the-art methods.**

***Index Terms*—Region-guided feature learning, remote sensing (RS) scene recognition, transferable visual prompting, unsupervised domain adaptation (UDA).**

I. Introduction

SCENE classification is a key research direction in the field of remote sensing (RS), aiming to automatically identify and classify different surface scenes by analyzing and processing image data acquired from RS devices [1]. This technology has been widely applied in various domains, including geographic information systems [2], environmental monitoring [3], urban planning [4], and agricultural management [5], providing scientific evidence and support for decision-making. However, traditional image recognition models often rely on large amounts of manually labeled data, which is costly and difficult to obtain. In addition, these models tend to have weak domain adaptation abilities, leading to poor performance in cross-domain tasks [6], [7]. To tackle this challenge, unsupervised domain adaptation (UDA) methods have been proposed [8], which help models effectively learn in target domains using data from source domains, alleviating the problem of insufficient labeled data. As a result, UDA has gradually become an important research direction in the field of RS image classification.

UDA refers to learning effectively in the target domain without labeled target-domain data, relying on data from the source domain to help the model adapt. Its primary objective is to minimize the distribution gap between the source and target domains, thereby improving the model's generalization capability on the target domain [9], [10]. Current research methods primarily include distribution discrepancy minimization [11], [12] and adversarial learning approaches [8], [13], [14], [15]. The distribution discrepancy minimization methods aim to reduce the distribution difference between the source and target domains by mapping features or designing loss functions, making the features from both domains as close as possible in a certain space. Adversarial learning methods, inspired by generative adversarial networks (GANs) [16], introduce an adversarial training process that enables the model to extract domain-invariant features automatically. However, the performance of these methods is highly dependent on prior domain knowledge, leading to poor robustness when handling noise and distribution differences in RS data, and in practical applications, performance degradation may occur [17].

In recent years, researchers have proposed domain adaptation methods specifically tailored for RS data to better address the complex environmental variations and noisy features in RS images. Zhu et al. [18] proposed a method that combines attention mechanisms with multiscale adaptive modules to effectively align the edge and conditional distributions, significantly improving cross-scene classification performance. In addition, Zhao et al. [19] proposed the SRKT method, which constructs an adaptive adversarial learning framework to align fine-grained conditional distributions and uses a semantic relationship transfer mechanism to reduce the discrepancy between source and target domains, further enhancing the model's generalization ability. On the contrary, Wang et al. [20] proposed the SERC method, which uses a distribution matching strategy to effectively mitigate domain shift

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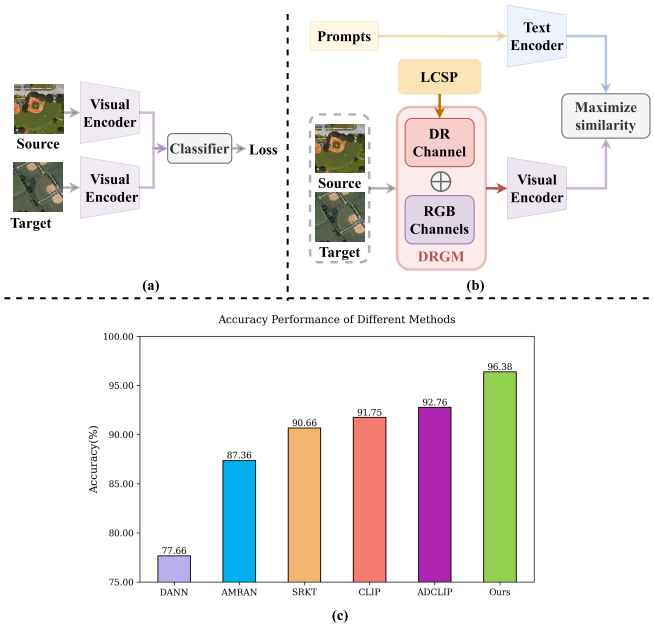


Fig. 1. Comparison between TDVP and existing methods. (a) Existing approaches primarily rely on single-modality learning, which constrains their generalization and robustness in domain adaptive scene recognition. (b) On the contrary, the proposed TDVP enhances recognition performance by integrating region-guided feature learning with cross-domain semantic prompting. (c) Accuracy comparisons on the UCM_WHU_AID_RSSCN7 datasets demonstrate the superior performance of TDVP across diverse benchmarks.

by leveraging the mutual reinforcement between coarse and fine classifiers. Although these methods have achieved some success in improving cross-scene classification performance and reducing domain differences, they still face challenges in enhancing generalization ability and robustness in complex environments. Especially in RS images, where semantic differences between scenes are large, traditional methods [21] often struggle to capture deep semantic information across different domains. As a result, recent research has started to explore the integration of vision-language pretraining models into UDA techniques for RS, aiming to better utilize cross-modal information for adaptive learning and further enhance the model's performance and stability, as illustrated in Fig. 1.

Visual-language models (VLMs) have recently demonstrated strong generalization capabilities in cross-domain tasks. Represented by CLIP [22], VLMs model the semantic associations between visual and textual modalities through joint learning of image and text representations, enabling zero-shot learning and cross-domain transfer. In addition, prompt learning [23], as a key technique in VLMs, guides the model to focus on specific semantic information by designing appropriate textual prompts, thereby enhancing task adaptability. To further optimize cross-domain adaptability, DAPL [24] embeds domain information into prompts, allowing the classifier to dynamically adapt to different domains. However, CLIP is prone to catastrophic forgetting during fine-tuning on the target domain, leading to a significant drop in accuracy. To address this, PADCLIP [25] introduces a catastrophic forgetting measure and adjusts the learning rate to prevent overfitting, effectively mitigating the forgetting problem. Moreover, the UniMoS [26] framework incorporates a modality separation network and modality ensemble training, successfully decoupling CLIP's visual and textual features while promoting cross-modal information interaction. In RS image classification, prompt learning helps VLMs attend to specific target features or scene information, thereby improving model adaptability and classification accuracy. Building on this, PADA-Net [49] integrates prompt information into an adversarial UDA framework, enhancing cross-modal alignment between images and text through an interactive bridging mechanism and dual metanets. Nevertheless, existing methods primarily focus on token-level alignment between text and image, neglecting structural information and discriminative features within images. This limitation is particularly evident in RS scene classification, where prompt-based domain adaptation methods show limited performance in handling spatially discriminative regions (DRs). To address this, we propose a transferable discriminative visual prompting (TDVP) approach for the RS scene domain adaptation, introducing a DR guidance module (DRGM) and a learnable cross-domain semantic prompt (LCSP) mechanism to align regional features with cross-domain semantics, thereby improving both cross-domain adaptability and classification performance.

This article presents an innovative framework for domain adaptive RS scene recognition via TDVP. The proposed method aims to address the limitations of existing prompt-based domain adaptation approaches in structural adaptability and spatially DR handling. In particular, the method incorporates the DRGM, which injects region-specific discriminative representations directly into the vision encoder, thereby enhancing the model's attention to and learning of key regions. This module precisely guides the model to identify and emphasize important regional features in the image, improving the model's feature extraction ability. Building upon this foundation, we further propose an LCSP mechanism, which uses a region-aware mask to generate spatially adaptive prompts, promoting semantic alignment between the source and target domains and significantly improving the model's transferability. To ensure semantic consistency across domains, we also introduce prompt consistency regularization (PCR), which minimizes the discrepancy between the image features of our model and the CLIP image features, thereby enhancing the stability and generalization ability of the model in cross-domain tasks.

The main contributions of this article are as follows.

1) To overcome the limitations of existing prompt-based domain adaptation methods in structural adaptability and spatial DR handling, we propose DRGM. This module injects region-specific discriminative representations into the vision encoder, effectively enhancing the model's attention to key regions and improving its performance in RS scene recognition.
2) Building on DRGM, we propose the LCSP mechanism, which uses a region-aware mask to generate spatially adaptive prompts, facilitating semantic alignment between the source and target domains, and enhancing the model's cross-domain transfer ability.

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3) To enhance the model's stability and generalization ability, we introduce PCR, which compares the semantic similarity between the image features of our model and the CLIP image features, ensuring consistent learning across domains.
4) Experimental evidence confirms that the proposed approach effectively mitigates domain shifts and enhances cross-domain performance, consistently surpassing existing advanced methods.

The remainder of this article is organized as follows. Section II presents a review of related work. Section III describes the proposed method. Section IV reports the experimental results and analysis. Section V concludes the article.

II. Related Works

Given that our research focuses on prompt-based scene recognition in the context of RS domain adaptation, Section II-A reviews the fundamental methodologies of UDA and its applications in RS scene recognition. Subsequently, Section II-B highlights recent advancements in the field of prompt learning. A UDA.

A. Unsupervised Domain Adaptation

Due to the difficulty of acquiring RS imagery and the high cost of annotation [6], [7], UDA has emerged as a research focus in recent years [16], [17]. The core objective of UDA is to mitigate the distribution discrepancy between a labeled source domain and an unlabeled target domain. Existing approaches can be broadly categorized into distribution alignment [11], [52] and adversarial learning [8], [16], [43], [44]. Distribution alignment methods reduce domain gaps through feature matching; for instance, DAN [11] employs mean matching in the reproducing kernel Hilbert space to enhance feature transferability. On the contrary, adversarial learning methods introduce adversarial mechanisms to learn domain-invariant representations. A representative example is the gradient reversal layer originally proposed by Ganin and Lempitsky [8], which enables models to simultaneously achieve discriminative ability and domain invariance during training, thereby significantly improving cross-domain generalization. Subsequent studies combined theoretical analyses to improve stability and interpretability, such as MDD [27], which defined a novel distribution discrepancy measure within the adversarial framework and provided generalization guarantees. With the growing adoption of Transformer architectures in computer vision, researchers have begun to explore their potential in UDA. CDTrans [28] was the first to introduce Transformer models and proposed a class-center-aware pseudolabeling strategy to improve the accuracy and stability of pseudolabels. PMTrans [29] further constructed intermediate domains between the source and target from a game-theoretic perspective and integrated MixUp strategies in both feature and label spaces, forming a tripartite adversarial framework to alleviate the domain alignment challenges of vision Transformers (ViTs) [48].

Although the aforementioned approaches have demonstrated strong performance in certain tasks, their applicability to cross-scene classification in RS imagery remains constrained due to the inherent complexity of semantic representation and domain distribution. Consequently, researchers have increasingly shifted their focus toward UDA methods tailored for RS data to improve cross-domain classification performance. In this context, Zheng et al. [30] proposed a cross-scene classification method that leverages target-domain class information and adversarial optimization to achieve precise source–target feature alignment. To further enhance generalization, Miao et al. [31] introduced the HFPAN framework, which constructs a hierarchical embedding structure with progressive alignment strategies to alleviate significant domain discrepancies. Moreover, Zhao et al. [19] proposed the SRKT model, which integrates adaptive adversarial learning with a semantic relation transfer mechanism to extract sensor-invariant features and reduce semantic structural gaps across classes, thereby improving model generalization. Distinct from these approaches, this article introduces a novel UDA framework based on transferable discriminative prompts to address domain shift in RS scene recognition. In particular, a DRGM is designed to strengthen the visual encoder's ability to attend to key regions, which is further integrated with cross-domain semantic prompting and region-aware masking to achieve more effective semantic alignment and feature transfer.

B. Prompt Learning

Benefiting from large-scale multimodal pretraining, VLMs have demonstrated remarkable generalization capabilities [22], [53], [54]. Among them, CLIP [22] stands out as a representative model, trained on approximately 400 million image–text pairs, and has shown exceptional performance in zero-shot image classification tasks. Inspired by the success of CLIP, prompt learning techniques have been widely integrated into VLM frameworks [24], [25], [26]. The central concept of prompt learning lies in enhancing the model's understanding and adaptability to downstream tasks by refining its language component, thereby improving overall performance. To address prompt engineering challenges in deploying pretrained VLMs such as CLIP, Zhou et al. [23] proposed context optimization (CoOp), which learns context tokens as trainable vectors while keeping model parameters frozen. Building on this idea, AD-CLIP [32] introduced a domain-agnostic strategy that enforces conditional constraints during prompt learning to obtain domain-invariant and class-generalizable prompts. PDA [33] further enhanced cross-domain recognition through a dual-branch prompt tuning architecture with interaction and joint optimization. From a task-decoupling perspective, DPC [34] employed a dynamic hard negative sampler to construct dual-prompt structures, balancing discriminability and generalization. To improve adaptability, VAMP [35] designed a vision-aware multimodal tuning mechanism, enabling visual prompts to guide text-specific prompt generation.

While these approaches have achieved strong results in general image classification, their direct transfer to RS imagery remains limited due to complex semantics and significant cross-scene distribution shifts. Recent efforts have therefore extended prompt learning to RS tasks. Zhu et al. [50] proposed MVP, integrating parameter-efficient tuning into a

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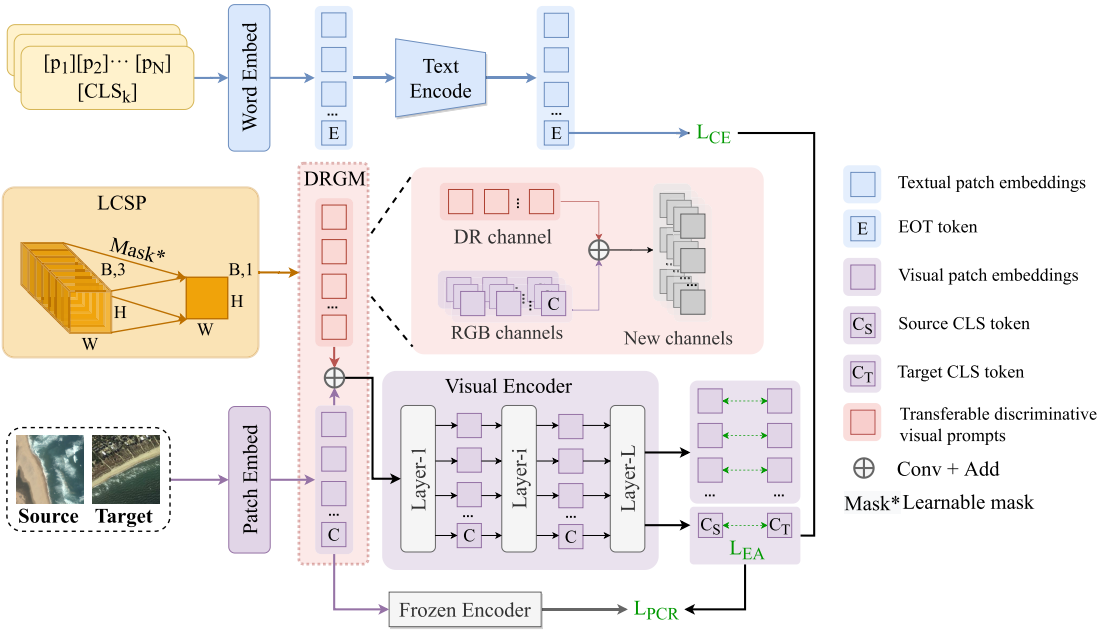


Fig. 2. Overview of the proposed TDVP framework. The framework consists of three key components: DRGM, LCSP, and PCR. DRGM first injects discriminative regional information into the visual encoder through an additional DR channel, enabling early stage attention to critical spatial regions. Based on the region-aware representation, LCSP then generates spatially adaptive prompts to facilitate semantic alignment between the source and target domains. Finally, PCR constrains the prompted features to remain semantically consistent with the frozen CLIP feature space, thereby improving the stability and generalization ability of the learned prompts across domains.

metalearning framework with patch-embedding recomposition for enhanced RS representation learning. SkySense V2 [51] adopted a unified Transformer backbone with modality-aware prompts, adaptive patch merging, and a mixture-of-experts module for efficient RS modeling. SenCLIP [52] transferred CLIP's pretrained representations to RS imagery with free-text prompts, significantly improving classification. For cross-domain scene recognition, PADA-Net incorporated prompt learning into adversarial domain adaptation, aligning visual–linguistic features via CLIP, an information bridge module, and a dual metanet structure. Building on these advances, we propose a region-aware prompting mechanism that introduces an auxiliary prompt channel to encode spatially DRs, guiding the model toward target-relevant areas and further improving cross-domain recognition.

III. Method

The proposed TDVP framework consists of three core modules that operate in a progressive manner. First, the DRGM introduces discriminative regional information into the visual encoding stage, enabling the model to focus on critical spatial regions at the early stage of feature extraction. Based on the resulting region-aware representation, the LCSP further generates spatially adaptive prompts to improve semantic alignment and transferability across domains. Finally, PCR constrains the prompted features to remain semantically consistent with the frozen CLIP feature space, thereby enhancing the stability and generalization ability of the learned prompts in cross-domain tasks. Fig. 2 displays the full architecture.

A. Preliminaries

In the UDA task, given labeled source domain $\mathcal{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and unlabeled target domain $\mathcal{D}_t = \{(x_i^t)\}_{i=1}^{N_t}$ where N_s and N_t represent the number of samples in the source and target domains, respectively. It is generally assumed that the two domains have different input distributions $P_s(X) \neq P_t(X)$, but share the same label space Y. To achieve cross-domain adaptation, we employ CLIP as the backbone network, leveraging its visual-language alignment capability for prompt-guided discriminative learning. CLIP contains an image encoder and a text encoder. The image encoder (e.g., ViT) extracts visual features v from input images, while the text encoder converts class-descriptive texts into embedding representations t_k. For each class k, the corresponding text prompt is defined as

$$t_k = [p_1][p_2]\dots[p_N][CLS_k] \tag{1}$$

where $[p_1][p_2]\dots[p_N] \in \mathbb{R}^{N\times D}$ denotes a learnable context vector of length N and dimension D and $[CLS_k]$ represents the class name of category k.

To achieve alignment between visual and textual representations within a unified feature space, classification is performed by computing the cosine similarity between image features and each class text feature. The probability is calculated as follows:

$$P(y=k|x) = \frac{\exp\left(cos\left(v, t_k\right)/\tau\right)}{\sum_{j=1}^{C}\exp\left(cos\left(v, t_j\right)/\tau\right)} \tag{2}$$

where τ is a temperature coefficient, $cos(,)$ denotes the cosine similarity, and C is the total number of classes.

B. DR Guidance Module

Traditional prompt-based methods usually introduce guidance information after feature extraction, which limits their ability to model discriminative regional cues at the early stage. To address this issue, we propose the DRGM, which injects

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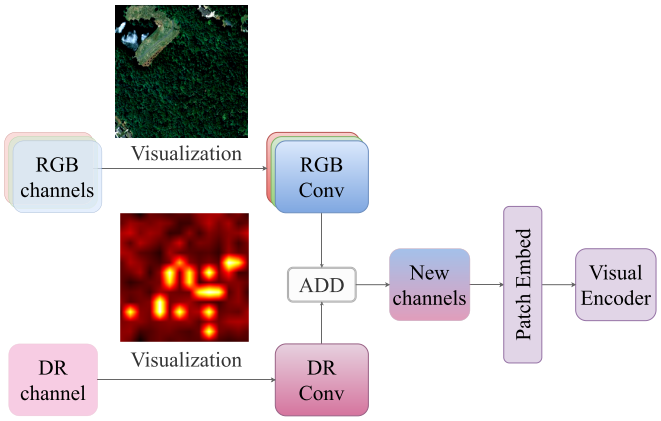


Fig. 3. Illustration of the proposed DRGM. The RGB channels and the DR channel are first processed by two parallel convolutional mappings, respectively. Their outputs are then fused to construct a new visual representation, which is subsequently fed into the visual encoder. In this way, discriminative regional information is explicitly incorporated into the early stage of feature extraction.

region-specific information directly into the visual encoder through a structural enhancement strategy. In particular, the original RGB image and the generated discriminative region mask are treated as two complementary inputs. The RGB channels and the DR channel are processed by two parallel convolutional mappings, respectively, and their outputs are fused before being fed into the visual encoder. In this way, DRGM enables discriminative regional cues to participate in early stage feature extraction and strengthens the model's robustness to domain shifts.

This process can be formulated as

$$x = \mathrm{Conv}_{\mathrm{rgb}}(RGB) + \mathrm{Conv}_{\mathrm{dr}}(DR) \tag{3}$$

where $x \in \mathbb{R}^{B\times 4\times H'\times W'}$ denotes the fused feature representation, $RGB \in \mathbb{R}^{B\times 3\times H\times W}$ is the original RGB channels, and $DR \in \mathbb{R}^{B\times 1\times H\times W}$ is the DR channel. $\mathrm{Conv}_{rgb}(\cdot)$ and $\mathrm{Conv}_{dr}(\cdot)$ represent convolutional mappings applied to the RGB and DR channels, respectively.

The structure of DRGM is illustrated in Fig. 3 via visualization. In particular, DRGM first introduces a DR mask generated from patch tokens as an additional DR channel, which is fused with the original RGB channels via a convolutional operation to form a four-channel input. On this basis, the patch-embedding layer of the image encoder is extended to process the fourth channel, thereby dynamically adjusting the model's focus on DRs at the early stage of feature modeling, strengthening the representation of target-specific features, and improving the overall discriminative capacity. By explicitly injecting salient-region information at the structural level, DRGM not only enables selective responses to critical regions but also enhances the model's robustness to domain shifts caused by variations in observation environments, thereby improving its cross-domain adaptability and generalization.

C. Learnable Cross-Domain Semantic Prompt

To enhance cross-domain semantic alignment and improve the transferability of image features, we propose a novel

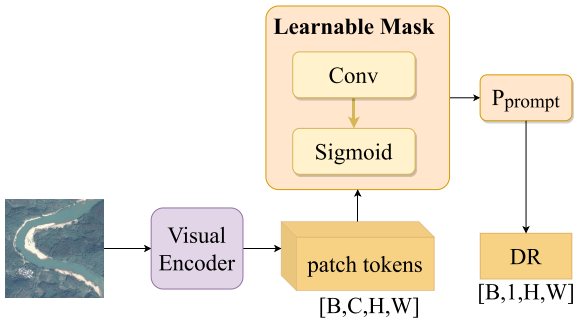


Fig. 4. Illustration of the LCSP process. A learnable region-aware mask dynamically generates cross-domain shared semantic prompts based on image features, which are then used to weight the image features, highlighting key regions and guiding feature learning.

mechanism based on DRGM—LCSP. LCSP uses a learnable region-aware mask to generate a set of cross-domain shared spatially adaptive prompts, which can adaptively capture the importance of different spatial locations based on the input image features. In addition, an embedding-level alignment loss is designed to ensure semantic consistency between the source and target domains. This mechanism not only achieves global feature alignment but also dynamically adjusts attention weights across regions, thereby better accommodating the diverse spatial structures of the target domain.

In particular, after the first forward pass of the visual encoder, the extracted patch tokens are reshaped into a 2-D feature map, as illustrated in Fig. 4. Let the patch-token representation be denoted as $F_p \in \mathbb{R}^{B\times C\times H_p\times W_p}$, where B is the batch size, C is the channel dimension, and $H_p \times W_p$ is the spatial resolution of the patch-token map. A learnable mask is then generated by applying a channel-compression convolution followed by a sigmoid activation:

$$M = \sigma\left(\mathrm{Conv}_{1\times 1}\left(F_p\right)\right) \tag{4}$$

where $\mathrm{Conv}_{1\times 1}(\cdot)$ denotes a 1×1 convolution that maps the patch-token feature map into a single-channel spatial response map and $\sigma(\cdot)$ is the sigmoid activation function. After that, the mask is upsampled to the input image resolution through bilinear interpolation

$$M^{\uparrow} = \mathrm{Interp}(M). \tag{5}$$

The resulting mask $M^{\uparrow} \in \mathbb{R}^{B\times 1\times H\times W}$ can be interpreted as a region-aware weighting map, where larger values indicate regions that are more important for cross-domain recognition. It is then used to modulate the visual encoding process, enabling the model to focus more on discriminative spatial regions and generate spatially adaptive semantic prompts. For the adopted ViT-B/16 backbone, the patch-token feature map is reshaped to 14×14, and the generated mask is upsampled to 224×224 before being fed back into the visual encoder.

To promote semantic consistency between the source and target domains, we introduce an embedding-level alignment loss to reduce the discrepancy between the prompted representations of the two domains. It should be noted that this loss is intended to encourage domain-level semantic alignment

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in the prompted feature space, rather than to indiscriminately force arbitrary samples to become similar. We employ an embedding-level alignment loss during training to measure the discrepancy between the weighted image features f^s_{masked} and f^t_{masked} from the source and target domains, respectively. This loss guides the learnable parameters of the learnable mask to generate cross-domain semantic prompts applicable to both domains. The specific computation is given by the following formula:

$$\mathcal{L}_{\mathrm{EA}} = \frac{1}{N}\sum_{i=1}^{N}\left\| f^s_{\mathrm{masked}} - f^t_{\mathrm{masked}} \right\|_2^2. \tag{6}$$

where f^s_{masked} and f^t_{masked} denote the learnable image features from the source and target domains, respectively; N is the number of samples; and $|\cdot|2^2$ represents the squared Euclidean distance used to measure the discrepancy between source and target image features. By introducing the embedding-level alignment loss $\mathcal{L}_{EA}$, the features from both domains are encouraged to maintain semantic consistency, enabling LCSP to learn a set of cross-domain shared semantic prompts, thereby effectively reducing distribution discrepancies in cross-domain tasks.

The extra overhead of TDVP mainly comes from the introduced DR channel and the lightweight mask generation in LCSP. Since the additional operations are limited to parallel channel processing, a 1×1 convolution, and bilinear interpolation, TDVP does not introduce an additional large backbone or heavy auxiliary branch.

D. Prompt Consistency Regularization

To enhance the stability and generalization of the learned prompts in cross-domain adaptation, we introduce a PCR strategy. The main idea is to preserve the semantic consistency between the visual features produced by our prompted model and the corresponding frozen CLIP features. In this way, the learned prompt-enhanced representations are constrained to remain in a stable semantic space, which alleviates prompt drift during cross-domain training.

In particular, given the prompted image features and the frozen CLIP features, we first apply ℓ_2 normalization to both feature sets. Their pairwise similarity matrix is then computed by normalized dot product, which is equivalent to cosine similarity. Based on this similarity matrix, we adopt a symmetric contrastive cross-entropy loss to encourage corresponding feature pairs within a minibatch to be distinguished from noncorresponding pairs.

For the source domain and the target domain, the similarity matrices are, respectively, defined as

$$S^S_{ij} = \frac{\hat{f}^C_i \cdot \hat{f}^S_j}{\tau} \tag{7}$$

$$S^T_{ij} = \frac{\hat{f}^C_i \cdot \hat{f}^T_j}{\tau} \tag{8}$$

where $\hat{f} = f/\|f\|_2$ denotes the ℓ_2-normalized feature and τ is the temperature coefficient.

Based on the above similarity matrices, the PCR loss for the source domain is defined as

$$L^S_{PCR} = \frac{1}{2}\left[\mathrm{CE}\left(S^S, y\right) + \mathrm{CE}\left(\left(S^S\right)^\top, y\right)\right] \tag{9}$$

where $y = [0, 1, \ldots, B-1]$ denotes the identity labels indicating diagonal positive pairs within a minibatch of size B.

Similarly, the PCR loss for the target domain is written as

$$L^T_{PCR} = \frac{1}{2}\left[\mathrm{CE}\left(S^T, y\right) + \mathrm{CE}\left(\left(S^T\right)^\top, y\right)\right]. \tag{10}$$

By introducing this symmetric contrastive consistency constraint, the source and target image features are both aligned with the frozen CLIP semantic space, thereby improving the stability and generalization ability of the learned prompts while preserving instance-level discrimination across domains.

In addition to PCR, we incorporate standard cross-entropy losses for both the source and target domains to provide direct supervision and further enhance cross-domain alignment. For the source domain, the cross-entropy loss is defined as

$$\mathcal{L}^s_{\mathrm{CE}} = -\sum_i y^s_i \log \hat{y}^s_i \tag{11}$$

where y^s_i denotes the one-hot vector of the ground-truth label for the ith source domain sample, and $\hat{y}^s_i$ represents the predicted probability distribution of the model.

For the target domain, the cross-entropy loss is defined as

$$\mathcal{L}^t_{\mathrm{CE}} = -\mathbb{I}\left(\Pr\left(\hat{y}^t_i\right) \geq h\right)\sum_i \hat{y}^t_i \log \tilde{y}^t_i \tag{12}$$

where $\hat{y}^t_i$ denotes the one-hot vector of the pseudolabel for the ith target-domain sample, $\tilde{y}^t_i$ is the model's predicted probability distribution, $\mathbb{I}(\cdot)$ is the indicator function, and h is a fixed confidence threshold used to select reliable pseudo-labeled target samples.

Finally, the overall loss function integrates all components as follows:

$$\mathcal{L}_{\mathrm{all}} = \mathcal{L}^s_{\mathrm{CE}} + \mathcal{L}^t_{\mathrm{CE}} + \alpha\mathcal{L}_{EA} + \beta\left(\mathcal{L}^S_{PCR} + \mathcal{L}^T_{PCR}\right) \tag{13}$$

where α and β are hyperparameters controlling the relative contributions of the $\mathcal{L}_{EA}$ and the $\mathcal{L}^S_{PCR}, \mathcal{L}^T_{PCR}$.

IV. Experiment

This section details the RS datasets and specific experimental configurations used in the study, and provides a comprehensive explanation and in-depth analysis of the experimental results by combining quantitative metrics with qualitative analysis.

A. Dataset

To comprehensively evaluate the effectiveness of the TDVP method, this study selected six mainstream benchmark datasets for RS scene classification for experimental validation.

1) *UCM [36]:* UCM is a classic high-resolution land-use dataset released in 2010, primarily designed for RS classification tasks. It includes 21 land-use categories with 100 images per category at a resolution of 256×256 pixels, amounting to 2100 images in total.

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The dataset features a spatial resolution of approximately 0.3 m.

2) *WHU-RS19 (WHU) [37]:* WHU is a high-resolution RS image dataset released in 2011, primarily designed for scene classification and object retrieval tasks. The dataset contains 19 categories of typical land-cover scenes, totaling 1005 images. All images were sourced from Google Earth, each having a resolution of 0.5 m and dimensions of 600 × 600 pixels.
3) *AID [38]:* AID is a high-resolution aerial imagery dataset released in 2017, primarily designed for RS scene classification tasks. The dataset consists of 30 categories of typical land-cover scenes, with each category containing 200–400 images, totaling approximately 10 000 images of 600 × 600 pixels. The images have a spatial resolution varying between 0.5 and 0.8 m.
4) *RSSCN7 [39]:* RSSCN7 is a high-resolution RS dataset for scene classification, initially released in 2015. It comprises seven categories of typical land-cover scenes, with 400 images per category, totaling 2800 RS images of 400 × 400 pixels.
5) *NWPU-RESISC45 (NWPU) [40]:* NWPU is a widely adopted large-scale dataset for RS scene classification, released in 2017. The dataset contains 45 categories of typical land-cover scenes. Each category contains 700 images, resulting in a total of 31 500 RGB images with a resolution of 256 × 256 pixels, and the spatial resolution varies from 0.2 to 30 m.
6) *PatternNet (PATTERN) [41]:* Released in 2018, PATTERN is a large-scale benchmark dataset tailored for RS image retrieval (RSIR) applications. The dataset comprises 38 land-cover scene categories, each containing 800 images, totaling 30 400 high-resolution RS images. The spatial resolution ranges from 0.06 to 4.69 m.

B. Experiment Setting

To validate the effectiveness of the proposed method, we selected common scene categories from six RS datasets and constructed three new dataset groups for cross-domain recognition tasks as detailed in Table I. The specific configurations are as follows.

First, four datasets were employed to construct 12 cross-domain transfer tasks involving a subset of shared categories. Second, three datasets were selected to build six bidirectional transfer tasks, each defined by different shared categories. Finally, another three datasets were used to form six transfer tasks based on ten shared categories.

We employed the ViT-B/16 architecture as the visual encoder with the number of context tokens set to 16, and utilized a pretrained CLIP text encoder. During training, the SGD optimizer was used with an initial learning rate of 0.002 for all datasets, dynamically adjusted by a cosine annealing scheduler [42]. The training process consisted of 30 epochs, with a fixed batch size of 16 per domain. The loss function hyperparameters were set to $\alpha = 0.1$ and $\beta = 1.0$. The confidence threshold in (12) was fixed to $h = 0.5$. All experiments were conducted on an NVIDIA GeForce RTX 3060 GPU with 12 GB of memory, and the implementation was based on the PyTorch framework using the Python programming language.

TABLE I
DATASET INFORMATION

Datasets	Instances	Classes	Tasks
UCM	700	6	U→W U→A U→R
AID	2220	6	A→U A→W A→R
WHU	316	6	W→U W→A W→R
RSSCN7	2400	6	R→U R→W R→A
UCM	1300	13	U→A
AID	4510	13	A→U
AID	6280	19	A→N
NWPU	13300	19	N→A
UCM	2000	20	U→N
NWPU	14000	20	N→U
AID	4870	10	A→P A→N
PATTERN	8800	10	P→A P→N
NWPU	8400	10	N→A N→P

C. Experiment Results

We compare TDVP with other state-of-the-art UDA and prompt learning methods, including ResNet50-based approaches: DAN [11], DANN [8], JAN [14], MCD [15], DDC [43], SWD [44], DTA [45], DuEDL [46], and Deep CORAL [47], as well as methods specifically designed for RS scenarios: DATSNET [30], AMRAN [18], HFPAN [19], and SRKT [31]. In addition, we evaluate ViT-based methods: CLIP [22], DAPL [24], ADCLIP [32], PDA [33], UniMoS [26], and the RS-specific PADA-Net [49]. The classification performance of all these methods is evaluated across three types of datasets. For clarity, the best and second-best results are highlighted in bold and underlined, respectively. By default, ResNet-based methods adopt the ResNet50 architecture, while ViT-based methods use the ViT-B/16 backbone.

Table II presents the experimental results for the UCM_WHU_AID_RSSCN7 datasets, comparing the performance of 13 adaptive classification methods based on ResNet and ViT architectures. The results show that with the ResNet50 backbone, TDVP improved classification accuracy by 5.72% over the best-performing SRKT method, while with the ViT backbone, TDVP outperformed the leading PADA-Net method by 1.33%. Overall, TDVP achieved an average accuracy of 96.38%, demonstrating its superior effectiveness in cross-domain adaptation tasks.

Table III presents the experimental results on the AID_UCM_NWPU datasets. In six cross-domain transfer tasks, our method consistently achieved the best or second-best accuracy, with an average accuracy of 93.22%, outperforming all compared approaches. With the ResNet backbone, TDVP improved over SRKT by 2.07%, and with the ViT backbone,

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TABLE II
RECOGNITION RESULTS (%) FOR THE UCM_WHU_AID_RSSCN7 DATASETS

backbone	Method/Task	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	average
ResNet	DAN [11]	74.68	67.07	57.50	80.14	92.79	72.79	80.29	99.37	76.88	82.86	88.92	85.32	79.88
	DANN [8]	73.51	66.39	57.24	79.65	88.67	72.18	75.53	96.68	72.39	80.31	87.96	81.36	77.66
	JAN [14]	76.58	69.19	61.89	81.86	92.29	73.85	82.86	98.10	78.13	81.57	88.61	84.28	80.75
	MCD [15]	81.60	78.73	64.52	85.42	95.11	76.64	82.71	99.40	80.55	86.62	93.41	92.63	84.78
	AMRAN [18]	89.56	84.80	69.02	85.20	94.82	79.50	88.16	99.68	81.00	87.56	94.59	94.48	87.36
	SRKT [31]	93.98	95.27	74.53	88.34	97.01	85.55	84.83	98.73	85.34	89.17	96.77	98.41	90.66
ViT	Source_only	94.70	95.00	84.30	90.70	97.50	86.50	92.20	99.00	85.60	95.00	97.50	94.00	92.67
	CLIP [22]	94.74	93.69	64.86	79.44	93.69	64.86	79.44	94.74	64.86	79.44	94.74	93.69	83.18
	DAPL [24]	96.20	95.99	78.25	80.50	95.95	77.08	79.83	95.57	76.33	78.00	86.71	86.89	85.61
	ADCLIP [32]	97.80	96.37	82.11	92.68	98.47	87.10	89.00	95.85	83.22	94.99	99.05	96.42	92.76
	PDA [33]	97.15	95.07	79.33	87.33	97.43	81.08	83.67	99.37	78.25	86.17	97.47	95.99	89.86
	UniMoS [26]	97.47	96.31	89.50	87.50	96.94	86.96	78.33	99.05	86.00	91.00	95.89	96.94	91.82
	PADA-Net [49]	96.84	95.90	88.04	**95.83**	96.80	87.79	**97.00**	**100.00**	87.33	**98.33**	98.73	97.97	95.05
	Ours	**99.00**	**98.60**	**90.20**	95.50	**99.20**	**91.70**	96.20	99.70	**91.00**	97.00	**99.70**	**98.80**	**96.38**

TABLE III
RECOGNITION RESULTS (%) FOR THE AID_UCM_NWPU DATASETS

backbone	Method/Task	A→U	U→A	N→U	U→N	N→A	A→N	average
ResNet	DAN [11]	75.40	56.63	85.27	54.15	86.66	86.72	74.14
	DANN [8]	67.21	52.84	84.40	53.31	86.08	82.27	71.02
	DDC [43]	68.38	52.84	80.47	52.12	85.30	81.31	70.07
	Deep CORAL [47]	65.69	53.94	79.75	52.21	85.10	81.40	69.68
	DATSNET [30]	85.67	88.40	91.87	77.53	94.33	88.40	87.70
	SRKT [31]	**95.38**	91.35	91.75	86.40	94.97	87.04	91.15
ViT	Source_only	84.40	89.90	89.90	84.50	91.50	86.70	87.82
	CLIP [22]	82.31	91.69	89.5	84.31	94.44	88.89	88.52
	DAPL [24]	70.31	75.43	65.10	63.80	66.21	58.95	66.63
	ADCLIP [32]	83.00	87.00	88.60	87.70	93.90	87.90	88.02
	PDA [33]	84.62	91.95	75.45	72.56	72.13	71.44	78.03
	UniMoS [26]	69.67	85.18	85.70	**90.89**	43.91	59.33	72.45
	PADA-Net [49]	92.20	96.10	92.10	88.60	94.30	88.20	91.92
	Ours	93.70	**97.30**	**93.40**	89.10	**95.90**	**89.90**	**93.22**

it exceeded PADA-Net by 1.30%, demonstrating its strong capability in cross-modal domain adaptation tasks.

Table IV presents the experimental results on the AID_NWPU_PATTERN datasets. Our method achieves the highest accuracy across all six cross-domain transfer tasks, with an average accuracy of 98.55%, outperforming all compared approaches. With the ResNet backbone, TDVP improves over HFPAN by 8.42%, and with the ViT backbone, it exceeds PADA-Net by 4.31%, demonstrating its effectiveness in mitigating domain discrepancies and enhancing model generalization.

D. Empirical Analysis

This section provides a comprehensive empirical analysis to validate the effectiveness of the proposed TDVP framework.

1) Ablation Study: To systematically evaluate the contribution of each component in TDVP, we perform ablation experiments in a one-factor-removal manner. In particular, each variant is constructed by removing one component from the complete model while keeping all the remaining components unchanged. Therefore, each ablation result reflects the contribution of the removed component under the joint effect of the other components. The evaluated variants include removing the embedding-level alignment loss ($w/o\ L_{EA}$), removing the PCR term ($w/o\ L_{PCR}$), removing the DR channel ($w/o\ DR$), and removing the learnable mask ($w/o\ MASK^*$). In particular, $w/o\ L_{PCR}$ denotes the variant obtained by removing the PCR term from the full TDVP framework.

Table V presents the results of the ablation study, clearly demonstrating the critical role of each component in domain adaptation. The baseline model trained solely on source domain data shows limited performance. In the complete model, removing the embedding-level alignment loss leads to a 0.78% drop in accuracy while eliminating the PCR loss results in a 2.15% reduction, indicating its significant contribution to classification performance. In addition, excluding the DR channel causes a 3.07% decline, underscoring its importance in guiding the model to focus on DRs and enhancing feature representations. Similarly, removing the learnable mask decreases performance by 2.20%, highlighting its indispensable role in improving spatial alignment and adaptability. Overall, these findings not only validate the independent contributions of each component in mitigating domain shift but also reveal their complementary and synergistic effects, enabling the complete model to achieve a 3.71% improvement over the baseline and significantly enhancing generalization and robustness in cross-domain tasks.

2) Comparison of Learnable Mask Strategies: To validate the effectiveness of the learnable mask method proposed in this article, we conducted a series of comparative experiments, evaluating it against a simple linear layer (Linear), a pseudolabel-guided (PLG) approach, and an MLP-based method (MLP). As shown in Table VI, our method achieved

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TABLE IV
RECOGNITION RESULTS (%) FOR THE AID_NWPU_PATTERN DATASETS

backbone	Method/Task	A→P	A→N	P→A	P→N	N→A	N→P	average
ResNet	DAN [10]	82.60	84.20	66.50	69.30	89.00	92.20	80.63
	DANN [8]	90.10	89.40	77.90	75.60	92.70	91.00	86.12
	JAN [14]	91.00	88.10	77.00	79.80	89.40	92.00	96.22
	MCD [15]	67.42	87.71	73.49	71.06	96.14	91.47	81.19
	AMRAN [18]	95.10	94.90	72.60	70.50	94.80	93.70	86.93
	HFPAN [19]	97.80	95.80	78.30	77.30	97.50	94.10	90.13
ViT	Source_only	95.50	91.90	90.20	87.80	92.30	94.10	91.97
	CLIP [22]	93.20	73.80	83.20	73.80	83.20	93.20	83.40
	DAPL [24]	86.80	68.90	80.50	69.60	80.10	87.10	78.83
	ADCLIP [32]	96.40	89.40	92.50	88.70	93.60	95.40	92.67
	PDA [33]	93.70	94.30	90.70	91.70	96.80	92.90	93.35
	UniMoS [26]	87.70	52.50	71.20	56.90	64.30	72.50	67.52
	PADA-Net [49]	99.59	91.42	96.47	80.93	97.21	99.81	94.24
	Ours	**99.70**	**96.60**	**99.20**	**97.00**	**99.30**	**99.50**	**98.55**

TABLE V
ABLATION STUDY OF TDVP (UCM_WHU_AID_RSSCN7)

Component/Task	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	average
Source_only	94.70	95.00	84.30	90.70	97.50	86.50	92.20	99.00	85.60	95.00	97.50	94.00	92.67
w/o $\mathcal{L}_{EA}$	98.70	98.40	88.80	93.70	99.10	89.40	96.00	99.10	89.60	96.20	**99.70**	98.50	95.60
w/o $\mathcal{L}_{PCR}$	98.70	98.20	87.80	87.20	99.10	84.10	92.80	99.40	88.60	96.50	**99.70**	98.70	94.23
w/o DR	98.10	97.70	89.30	90.30	98.80	86.30	82.70	99.40	83.60	95.70	99.40	98.40	93.31
w/o $MASK^*$	98.10	97.90	**90.30**	88.80	98.90	84.00	95.70	99.00	84.00	95.70	99.40	98.40	94.18
Ours	**99.00**	**98.60**	90.20	**95.50**	**99.20**	**91.70**	**96.20**	**99.70**	**91.00**	**97.00**	**99.70**	**98.80**	**96.38**

TABLE VI
COMPARISON OF LEARNABLE MASK METHODS (UCM_WHU_AID_RSSCN7)

Method/Task	U→W	U→A	U→R	W→U	W→A	W→R	A→U	A→W	A→R	R→U	R→W	R→A	average
$Linear$	98.10	97.30	89.90	92.50	98.60	85.10	82.20	99.40	89.10	95.30	99.10	98.20	93.73
PLG	98.40	98.50	89.50	94.70	98.40	82.50	95.70	**99.70**	81.90	95.50	98.70	98.30	94.32
MLP	98.10	97.80	88.50	94.50	98.70	88.80	90.20	99.40	84.20	95.80	99.00	98.00	94.42
$Conv$(Ours)	**99.00**	**98.60**	**90.20**	**95.50**	**99.20**	**91.70**	**96.20**	**99.70**	**91.00**	**97.00**	**99.70**	**98.80**	**96.38**

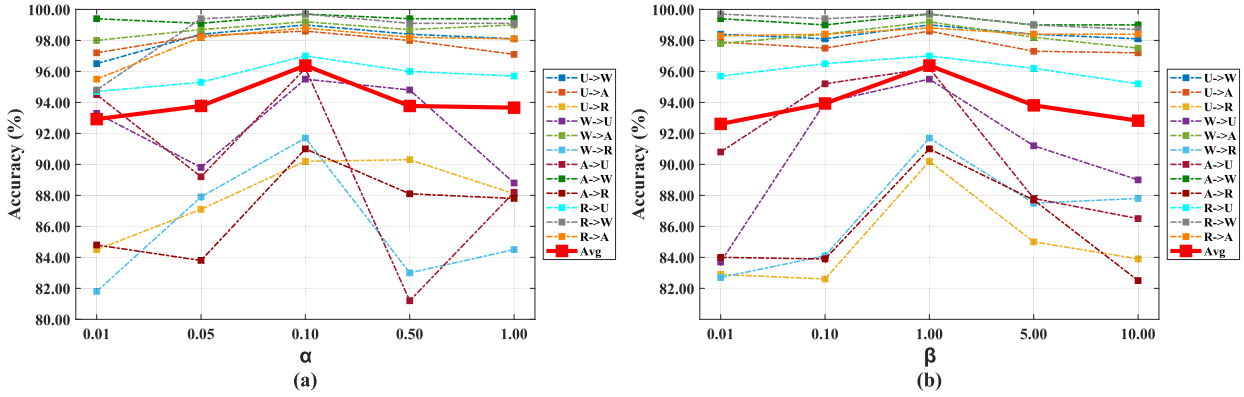


Fig. 5. Hyperparameter sensitivity analysis. (a) Performance variation with respect to the embedding-level consistency loss weight α. (b) Performance variation with respect to the PCR loss weight β. Optimal performance is achieved at $\alpha = 0.10$ and $\beta = 1.00$.

significant improvements across all 12 transfer tasks, demonstrating stronger feature learning capability and adaptability, ultimately attaining the best overall performance. These results fully confirm the superiority of the learnable mask method in enhancing model effectiveness.

3) Sensitivity Analysis of Parameters: A hyperparameter sensitivity analysis was conducted on the UCM_WHU_AID_RSSCN7 dataset, focusing on the impact of the embedding-level consistency loss weight α and the PCR loss weights β. As illustrated in Fig. 5, the model achieves peak performance at $\alpha = 0.10$ within the range of 0.01 to 1.00. Similarly, testing β within the range of 0.01 to 10.00 indicates optimal results at $\beta = 1.00$. These findings validate the appropriateness of setting α to 0.10 and β to 1.00.

4) Learnable Prompt Length: As illustrated in Fig. 6, the experimental results on the UCM_WHU_AID_RSSCN7 dataset indicate that when the learnable prompt length N is set to 4, 8, 16, 32, and 64, the model achieves optimal performance at N = 16. Considering the significant impact of this

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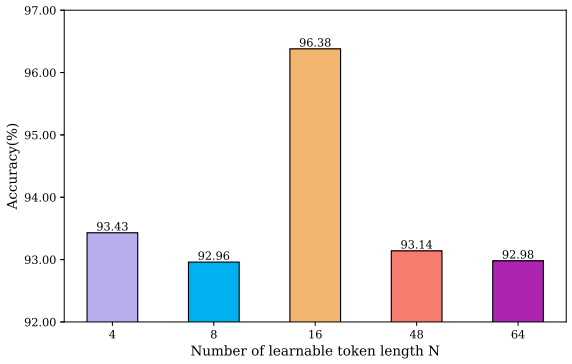


Fig. 6. Effect of different learnable prompt lengths on model performance. Optimal performance is achieved at $N = 16$.

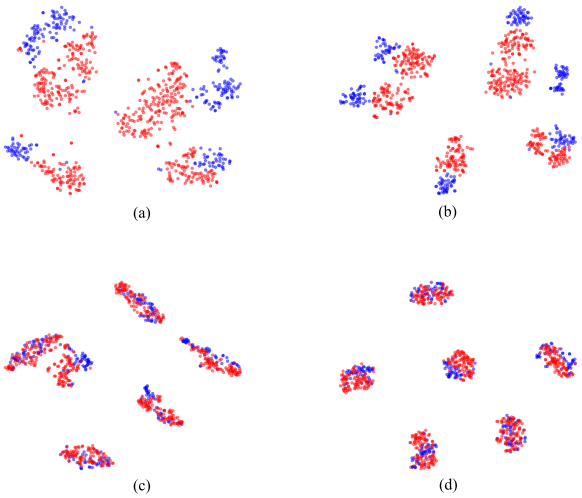


Fig. 7. Perform t-SNE visualization on the W→U task of the dataset group I. (a) DANN. (b) PDA. (c) ADCLIP. (d) TDVP(ours).

parameter on model performance, we fix the learnable prompt length at 16 to fully leverage the model's expressive power and adaptability, ensuring high accuracy while efficiently utilizing computational resources.

5) Features Visualization: To evaluate feature alignment, we compared the feature distributions of TDVP with DANN, PDA, and ADCLIP on the W→U task of the UCM_WHU_AID_RSSCN7 dataset. The t-SNE visualizations in Fig. 7 demonstrate that TDVP achieves superior alignment between source and target-domain features, with clearer interclass boundaries compared to other methods. These visual results further confirm the effectiveness of TDVP in cross-domain feature alignment and class discrimination.

The cross-domain alignment effect of TDVP is mainly reflected by the transfer results on multiple tasks, the ablation study, and the feature visualization in Fig. 7. These results consistently indicate that the proposed method reduces domain discrepancy while improving class discrimination. Although TDVP achieves the best average performance across all three dataset groups, it does not rank first on every individual transfer task. In particular, on several relatively easier transfer directions in dataset group I, such as W→U, A→W, and R→U, strong prompt-based baselines achieve slightly better results. We attribute this phenomenon to two factors. First, when the domain discrepancy is relatively limited, or the scene structure is comparatively simple, strong baselines can already obtain highly competitive performance, leaving less room for further improvement. Second, the main advantage of TDVP lies in exploiting discriminative regional cues and region-aware semantic alignment, which is particularly beneficial under larger domain shifts or more challenging cross-domain settings. Therefore, for transfer tasks dominated by relatively stable global scene layouts, the gain brought by explicit regional enhancement may be less significant. This observation also reflects a limitation of the proposed method, namely that its superiority is more pronounced on challenging transfer scenarios than on relatively simple ones.

V. Conclusion

The task of RS scene recognition is challenged by severe domain distribution shifts caused by complex and variable observation environments, making UDA a promising solution. To address the limitations of existing prompt-based approaches, which often overlook structural adaptability and spatially discriminative cues, we present a transferable visual prompting framework that explicitly incorporates region-specific information into the early stages of feature extraction. The framework further employs spatially adaptive prompting to align semantics across domains and introduces a consistency constraint to stabilize learned representations. This design enables the model to focus on critical regions, reduce cross-domain discrepancies, and enhance generalization. Extensive experiments on multiple public datasets validate the effectiveness of the proposed approach, consistently surpassing state-of-the-art methods in mitigating domain shifts. Despite the promising results, TDVP still has certain limitations. Its advantage is more evident on challenging transfer tasks with larger domain discrepancy, while the gain may be less significant on relatively easier transfer settings. In addition, its effectiveness depends on the quality of the region-aware representation. When discriminative regional cues are weak, the improvement may be limited. Future research will explore extending this framework to more challenging settings, such as multisource adaptation and few-shot learning, as well as investigating more efficient strategies for prompt learning and cross-modal knowledge transfer. Although TDVP is evaluated on a CLIP-based framework in this work, its region-guided enhancement design is potentially applicable to other vision-language models or visual backbones with similar visual encoding structures. Exploring such extensions will be part of our future work.

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